

1. Project Proposal: Fake News Detection System using Python

Name: [Muqheet Ahmed (53102)]
[Abubakar Awan (54257)]

1. Introduction / Background

In the modern digital era, information spreads rapidly through social media and online platforms. However, not all information is reliable, and the spread of fake news has become a major issue. Fake news can mislead people, influence public opinion, and create confusion in society. With the advancement of Artificial Intelligence, it is possible to analyze textual data and automatically detect whether a news article is real or fake.

This project aims to develop a Fake News Detection System using Python that classifies news articles based on their authenticity using machine learning techniques.

2. Problem Statement

The problem is to design an intelligent system that can automatically detect whether a given news article is fake or real based on its textual content, helping users identify reliable information.

3. Objectives

- To build a machine learning model that classifies news as real or fake.
- To preprocess and clean textual data using NLP techniques.
- To apply TF-IDF for feature extraction.
- To train a classification model such as Naive Bayes or Logistic Regression.
- To develop a simple user interface for testing news input.

4. Methodology

- **Data Source** – Use a publicly available dataset such as Fake and Real News Dataset from Kaggle.
- **Data Preprocessing** – Clean the text by removing stopwords, punctuation, and irrelevant characters using NLP techniques.
- **Feature Extraction** – Convert text into numerical format using TF-IDF Vectorization.
- **Model Training** – Train a classification model such as Naive Bayes or Logistic Regression on the dataset.
- **Evaluation** – Evaluate model performance using accuracy, precision, and recall.

- **Prediction System** – Develop a system where users can input news text and get prediction results (Fake or Real).
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5. Expected Outcomes

- A working Fake News Detection System using machine learning.
 - Accurate classification of news articles as fake or real.
 - Better understanding of NLP and text classification techniques.
 - A simple and user-friendly interface for users.
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6. Tools and Technologies

- **Programming Language:** Python
 - **Libraries:** Pandas, NumPy, NLTK, Scikit-learn
 - **Technique:** TF-IDF, Naive Bayes / Logistic Regression
 - **Dataset:** Fake and Real News Dataset (Kaggle)
 - **Environment:** Jupyter Notebook or Google Colab
 - **Deployment:** Streamlit or Flask
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7. Timeline

- **Week 1:** Dataset collection and preprocessing
 - **Week 2:** Feature extraction and model training
 - **Week 3:** Model evaluation and improvement
 - **Week 4:** Deployment and report preparation
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8. References

- Kaggle Dataset – <https://www.kaggle.com/>
- Scikit-learn Documentation – <https://scikit-learn.org/>
- NLTK Documentation – <https://www.nltk.org/>
- Streamlit Documentation – <https://streamlit.io/>